

Information Technologies for Industrial Engineers

General Information

- **Course Number:** 255411
- **Course Name:** Information Technologies for Industrial Engineers
- **Semester:** 2569-1
- **Instructor:** Nirand Pisutha-Arnond

After this class you will be able to

- Describe the landscape of information technology.
 - Web technology (frontend, backend)
 - Artificial intelligence
 - Blockchain
- Develop information system for industrial organizations.
 - Web application
 - AI-powered web application
 - Decentralized application

Past-year apps

- Cloth-size prediction
- Image classification
- Object Detection
- Speech Recognition
- Question and answer
- Show my blockchain assets

วิธีการประเมินผลการเรียนรู้	สัดส่วนของการประเมิน
การสอบกลางภาค	20%
การสอบปลายภาค	20%
Project	60%

แผนการสอน

- Midterm (Web application)
 - HTML, CSS, JavaScript
 - Web framework (React JS)
 - Database
- Final (Empowering web application)
 - AI
 - Blockchain

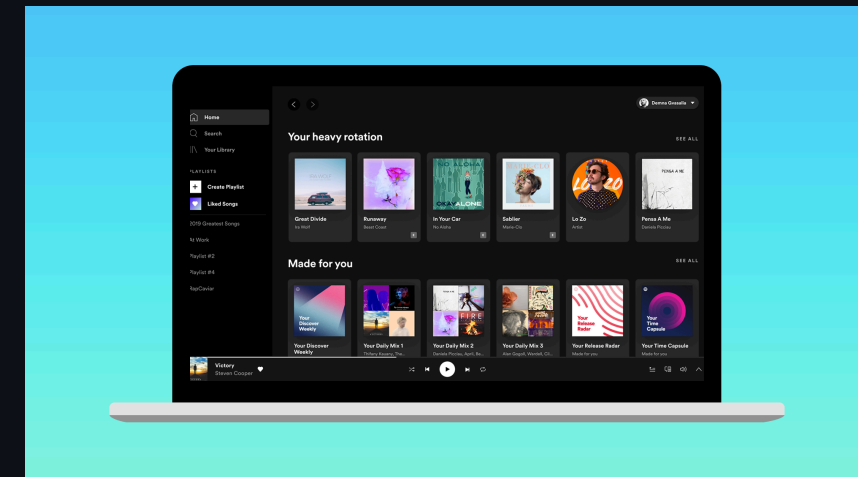
Mango Canvas Setup



Web Application...?

Software

- Applications that runs on top of an operating system
 - Windows, MacOS, Linux, Android, iOS
- "Talk" to OSs directly.
- Cannot be used between operation systems.
- Use different technologies/languages
 - C# (Windows), Swift (MacOS, iOS), Kotlin (Android), C++ (Linux)



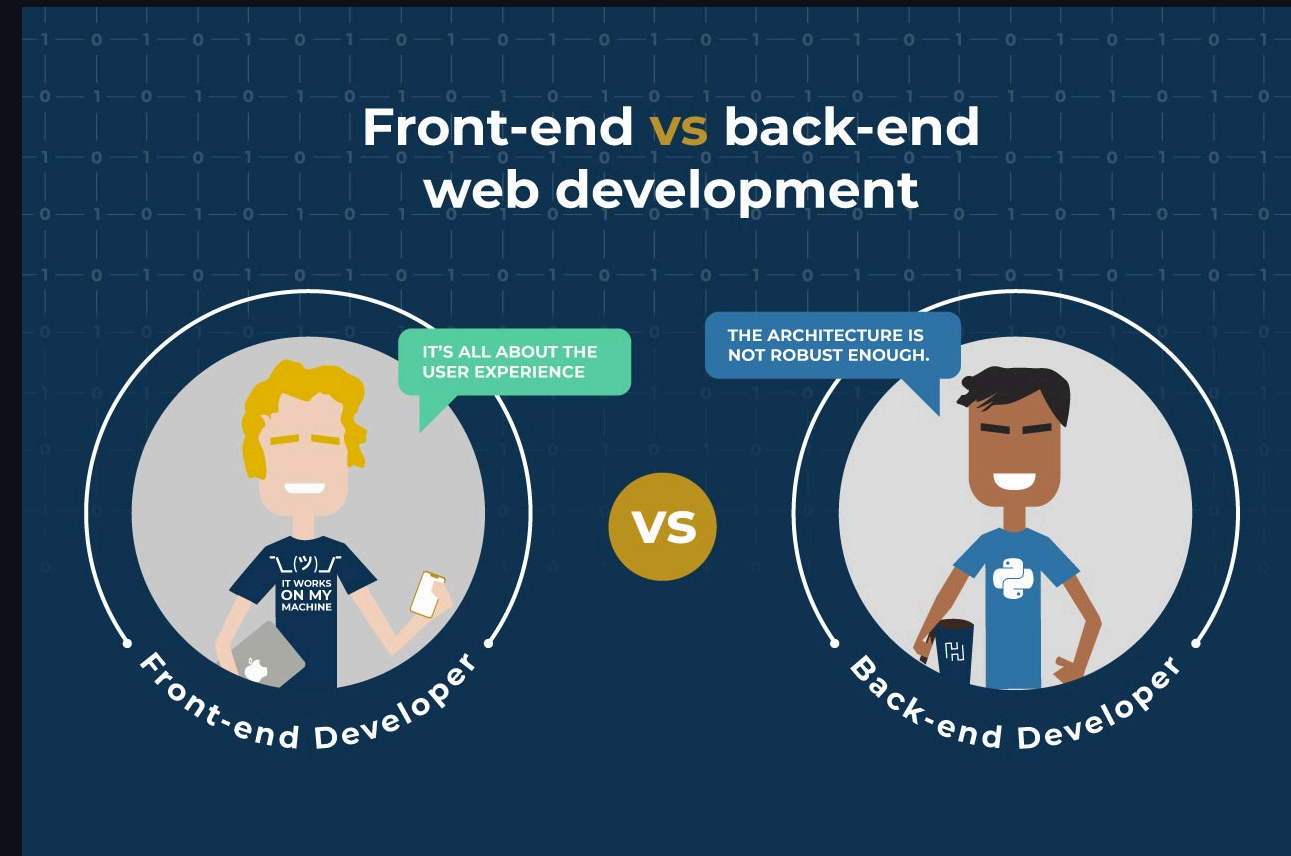
Web Application

- Applications that execute on a browser (another application) that runs on top of an operating system.
- Does not "talk" to the system directly.
- *Cross platform*
 - "Write once, use anywhere"
- Use similar technologies/languages
 - HTML, CSS, JavaScript



Types of "Web" Developers

- Front End
- Back End
- Dev Ops
- Full Stack



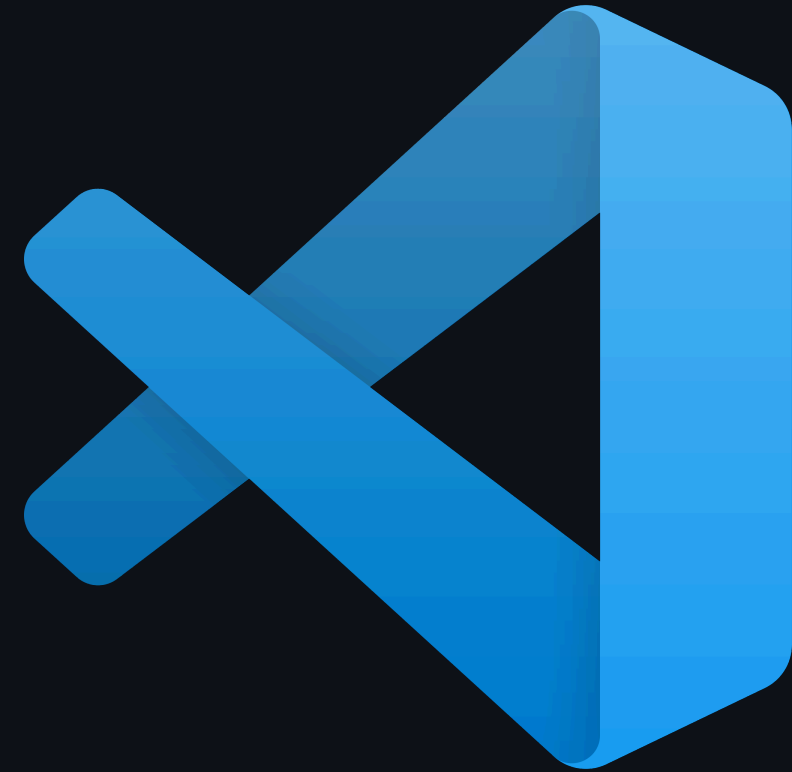
Web Developer Roadmap

Do not panic!

Lets get started

Tools

- Web browser
 - Chrome
- Integrated development environment (IDE)
 - VS Code
- *Let's go*



Recommended extensions/settings

- Enable: Format on save
- Extension: Prettier
- Extension: Auto Rename Tag